

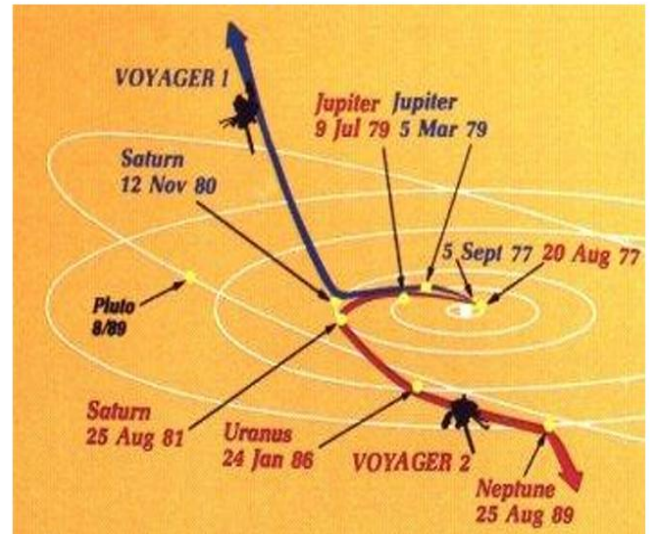
Question & Answers

1. Why does the visual geometry prove that gravity assists fundamentally redirect trajectories rather than merely accelerate the spacecraft, and what irreversible decision did this force upon Voyager 1 at Saturn?

Answer:

The flight trajectories diagram (page 1, left) visually shows Voyager 1 and Voyager 2 making sharp, angled direction changes at each planet — especially noticeable at Jupiter and Saturn. These bends prove that gravity assists are not just speed boosts; they fundamentally redirect the spacecraft's path by using the planet's gravity to swing it onto a new heading without expending fuel. The incoming and outgoing paths form clear angles, not a straight line with only velocity increase.

This redirection created an irreversible decision for Voyager 1. It was deliberately targeted for a close flyby of Titan (Saturn's large moon with a dense atmosphere), which changed its trajectory so dramatically that it was ejected out of the ecliptic plane and could never reach Uranus or Neptune. Voyager 2 avoided a close Titan pass, preserving the option for the extended Grand Tour to Uranus and Neptune.

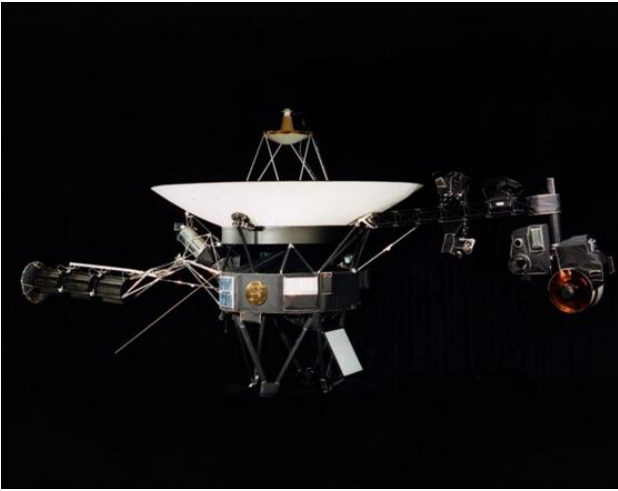


Page 1: “Voyager 1 would also be targeted to fly by Saturn’s moon Titan... a trajectory that would preclude any future planetary flybys.” Diagram visually shows Voyager 1’s path bending sharply and leaving the plane after Saturn/Titan.

2. Why was this photograph considered a major engineering milestone even though the spacecraft was still relatively close to Earth by solar system standards?

Answer:

The photograph (page 2, left) is the first time any spacecraft captured both Earth and the Moon together in a single frame, taken at 7.25 million miles from Earth only 13 days after launch. Even though 7.25 million miles is tiny compared to the solar system scale, it was a major engineering milestone because it proved the imaging system could handle extreme dynamic range (bright Moon against deep space), precise pointing accuracy, and reliable long-distance photography — all essential capabilities needed later for imaging distant planets, moons, and rings.



Page 2: “Just 13 days after its launch, Voyager 1 scored the first of its many firsts: at a distance of 7.25 million miles, it... snapped the first ever photograph of the Earth-Moon system in a single frame...”

3. What specific visual features in this montage support the claims of active volcanoes on Io and a possible subsurface ocean on Europa?

Answer:

In the montage on page 3 (left):

Io: The moon shows bright orange-red spots and dark plume-like features extending from the surface — clear visual signs of active volcanic eruptions and lava flows.

Europa: The moon appears very smooth, bright white, and crisscrossed with long linear cracks, but has very few visible impact craters — strong evidence of recent geological resurfacing, consistent with a liquid water ocean beneath the icy crust pushing up and smoothing the surface.



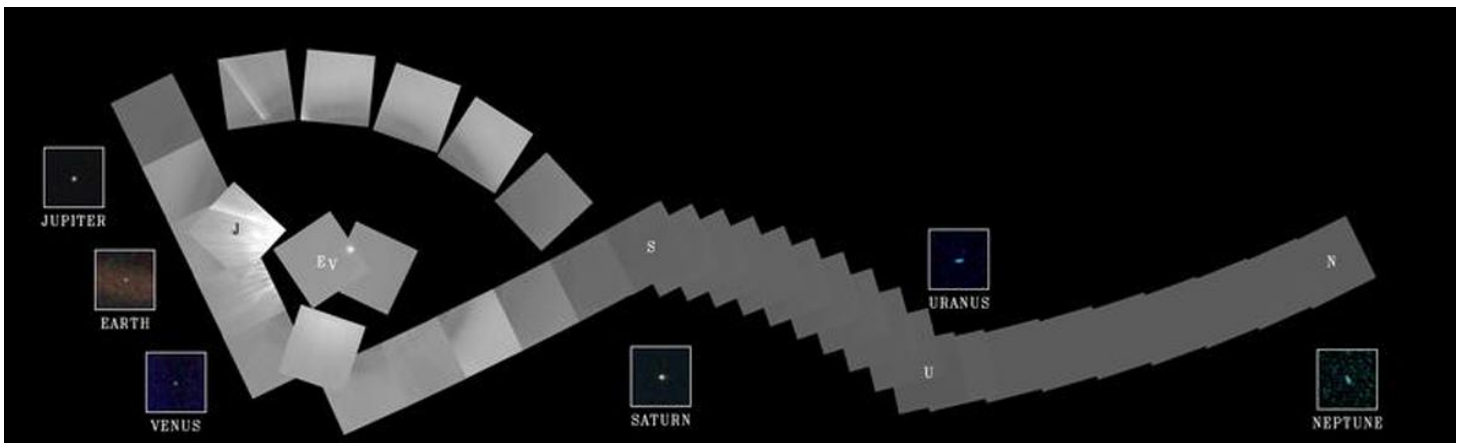
Page 3: “the moon Io has active volcanoes, the moon Europa has a geologically young surface made of ice possibly floating on an ocean of liquid water...”

4. Why did the mission team deliberately wait until just before permanently shutting down the cameras to capture this image, and what deeper message does it convey?

Answer:

The team deliberately waited until February 14, 1990 — after the final planetary encounter (Neptune in August 1989) and just before permanently turning off the cameras to conserve power — to take this mosaic of 60 images. The image shows Earth as a tiny pale blue dot among other planets and vast emptiness. The deeper message is philosophical: from 6 billion kilometers away, Earth appears insignificant, emphasizing humanity’s smallness in the cosmos and the unity of our shared planet.

Direct evidence from PDF:



Page 4: “On February 14, 1990... it was able to capture a ‘family portrait’ of six of the solar system’s planets, including a pale blue dot called Earth. It was fitting that these were the last pictures from either Voyager.”

5. Why was 1970s analog technology deliberately chosen for a message that might be found tens of thousands of years in the future?

Detailed Answer:

The Golden Record uses 1970s analog phonograph technology because it is extremely robust and requires only simple mechanical playback (a needle and turntable) — no electricity, computers, or digital decoding needed. Any civilization with basic mechanical understanding could play it. Digital formats might become obsolete or unreadable over tens of thousands of years, while analog grooves are physically durable.

Direct evidence from PDF:



Page 5: “each Voyager is equipped with a golden record (think 1970's technology!)... Just hope those aliens have turntables! Even a stylus was thoughtfully provided with the record, along with instructions on how to play it.”

6. What scientific trade-off led to Voyager 1's flyby permanently sacrificing all future planetary encounters while Voyager 2 was allowed to continue?

Detailed Answer:

The trade-off was prioritizing a close study of Titan's dense atmosphere (a high-priority scientific target) versus preserving the option for future flybys. Voyager 1 was sent on a close Titan flyby (4,000 miles), which dramatically altered its trajectory and ejected it out of the ecliptic plane, making Uranus and Neptune unreachable. Voyager 2 took a less optimal Saturn path, avoiding close Titan contact, so it could continue to Uranus and Neptune.

Direct evidence from PDF:



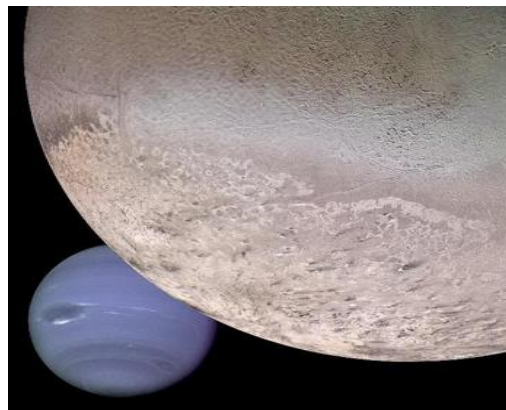
Page 3: "Voyager 1 made a close flyby of Saturn's large moon Titan... a trajectory that would preclude any future planetary encounters."

7. Why is it counterintuitive that Voyager 2 discovered more new moons at these distant ice giants than at Jupiter and Saturn combined, despite spending far less time observing them?

Detailed Answer:

Voyager 2 discovered 11 new moons at Uranus and 6 at Neptune (17 total), compared to only a few at Jupiter and Saturn, even though observation time at the ice giants was much shorter. This is counterintuitive because closer planets were studied longer, but the ice giants had many small, previously undetected moons that only became visible during close flybys — the distance and faintness made them hard to detect from afar.

Direct evidence from PDF:



Page 4: "discovering 11 new moons" (Uranus) and "discovering 6 new moons" (Neptune).

8. How does the visible design explain why both spacecraft could still be operating after 45+ years when most missions are planned for only a few years?

Detailed Answer:

The spacecraft illustration (page 2, right) shows large high-gain antennas, radioisotope thermoelectric generators (RTGs), and a robust, redundant design with minimal moving parts. The RTGs provide steady, long-lasting power from plutonium-238 decay (independent of distance from the Sun), unlike solar panels that degrade or become ineffective far from the Sun. This engineering allowed both Voyagers to continue operating far beyond their original Jupiter-Saturn mission.

Direct evidence from PDF:



Page 5: “Forty years after they were launched, both spacecraft are still operating... can continue to function until about 2025.”

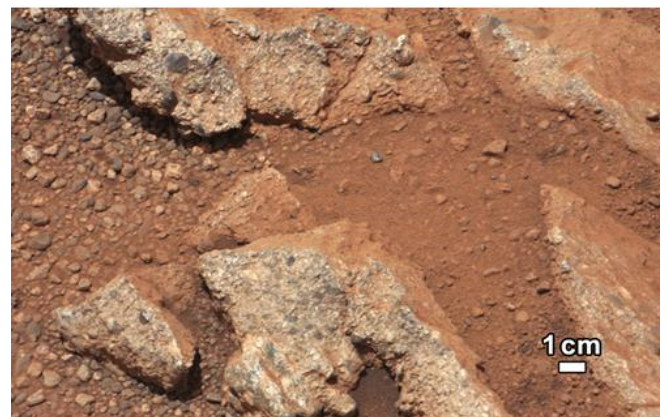
9. Reconstruct the full ancient river system story and explain why this image alone was not sufficient to prove past habitability.

Detailed Answer:

The Link outcrop image (page 2, top) shows rounded pebbles mixed with hardened sand in conglomerate rocks — clear evidence of ancient vigorous stream flow that rounded the pebbles and deposited them. This proves mechanical water activity on the surface. However, this single image was not sufficient to prove past habitability. It only shows water movement; later drilling at John Klein provided mineralogical evidence of sustained liquid water, key chemical ingredients for life, a chemical energy source, and water that was not too acidic or salty — all four factors needed together.

Direct evidence from PDF:

Page 2: “The evidence for stream flow was in rounded pebbles mixed with hardened sand in conglomerate rocks...” and later drill analysis.



10. Why was the 20 km landing precision improvement not just an engineering achievement but a fundamental prerequisite for the entire scientific mission?

Detailed Answer:

Previous Mars landings had uncertainty of ± 60 km or more. The sky crane image (page 1, top right) shows the precision descent system. Gale Crater's scientific target — the layered mountain Aeolis Mons ("Mount Sharp") — is located very close to the crater wall. Without the 20 km precision improvement, the landing would have been unsafe and the rover could not have reached the layered stratigraphy needed to study Mars' transition from a wetter past to a drier present.

Direct evidence from PDF:

Page 1: "landing within a target area only about 20 kilometers (12 miles) long... Gale Crater landing site is so close to the crater wall and Mount Sharp that it would not have been considered safe..."



11. How was the self-portrait image technically assembled, and what does it reveal about the rover's mobility and self-diagnostic capabilities?

Detailed Answer:

The self-portrait (page 3, top left) was constructed by the Mars Hand Lens Imager (MAHLI), a close-up camera mounted on the robotic arm. MAHLI captured dozens of individual images from different arm positions and angles, which were then stitched into a composite. This technique reveals the rover's remarkable arm articulation—the ability to position a camera at multiple vantage points around its own body. The absence of the arm itself in most of the final image demonstrates precise planning of camera angles. This capability serves dual purposes: public engagement and engineering diagnostics, allowing mission teams to visually inspect hardware conditions including wheel wear, dust accumulation, and instrument status.

Direct evidence from PDF:

Page 3: "Mounted on the arm, the Mars Hand Lens Imager takes extreme close-up pictures of rocks and soil, revealing details smaller than the width of a human hair. It can also focus on hard-to-reach objects more than an arm's length away and has taken images assembled into dramatic self-portraits of Curiosity."



12. How does the drilled sample from John Klein, combined with its 4.2 billion year age, create a paradox about geological preservation on Mars?

Detailed Answer:

The drilling image (page 2, bottom right) shows powder extracted from the John Klein rock's interior. Analysis revealed the rock formed 4.2 billion years ago—during Mars' early, wetter period—yet had only been exposed at the surface for 80 million years. This creates a preservation paradox: ancient habitable conditions were locked inside rock for over 4 billion years, protected from radiation and oxidation. The brief 80-million-year surface exposure means the evidence of past habitability remained pristine underground for 98% of Mars' history. This demonstrates that subsurface Mars may preserve biosignatures far better than the radiation-bombarded surface, fundamentally reshaping astrobiology search strategies toward drilling and subsurface exploration.

Direct evidence from PDF:



Page 2: "On a subsequent drill sample, Curiosity was able to accomplish a first for measurements on another planet: determining the age of the rock. The measurements showed that the drilled material was 4.2 billion years old and yet had been exposed at the surface for only 80 million years."

13. Why was the documented decline in radioisotope power from 110W to 100W over two years actually a positive indicator for long-term mission success?

Detailed Answer:

The radioisotope thermoelectric generator (RTG) uses plutonium-238 decay, which has a half-life of 87.7 years. The 9% power decline over two years closely matches the expected decay curve, confirming the system was performing precisely as designed with no unexpected degradation. More importantly, the document notes the generator was "still providing more than 100 watts two years after landing"—well above the minimum needed for operations. This predictable, gradual decline contrasts sharply with solar panels, which suffer unpredictable dust accumulation and dramatic seasonal variations. The steady power budget allowed mission planners to confidently extend operations beyond the 687-day primary mission, knowing exactly how much power would be available years into the future.

Direct evidence from PDF:



Page 3: "At launch, the generator provided about 110 watts of electrical power to operate the rover's instruments, robotic arm, wheels, computers and radio... Although the total power from the generator will decline over the course of the mission, it was still providing more than 100 watts two years after landing."

14. Why does the document argue that visual imaging alone could never have proven past habitability, and what hierarchical chain of instruments was required?

Detailed Answer:

Visual evidence from cameras (Mast Camera, Mars Hand Lens Imager) could identify sedimentary structures suggesting water flow, but couldn't prove chemical habitability. The hierarchical proof required: (1) ChemCam's laser spectroscopy to identify elemental composition at distance, (2) Sample collection by the robotic arm's drill, (3) CheMin's X-ray diffraction to identify clay minerals formed in neutral-pH water, (4) Sample Analysis at Mars to detect organic chemistry and measure isotope ratios indicating sustained liquid water, and (5) Alpha Particle X-ray Spectrometer to quantify key life elements (C, H, N, O, P, S). Only this complete chain—from remote sensing to molecular analysis—could demonstrate the five habitability requirements: liquid water, chemical building blocks, energy sources, appropriate pH, and preservation potential.

Direct evidence from PDF:

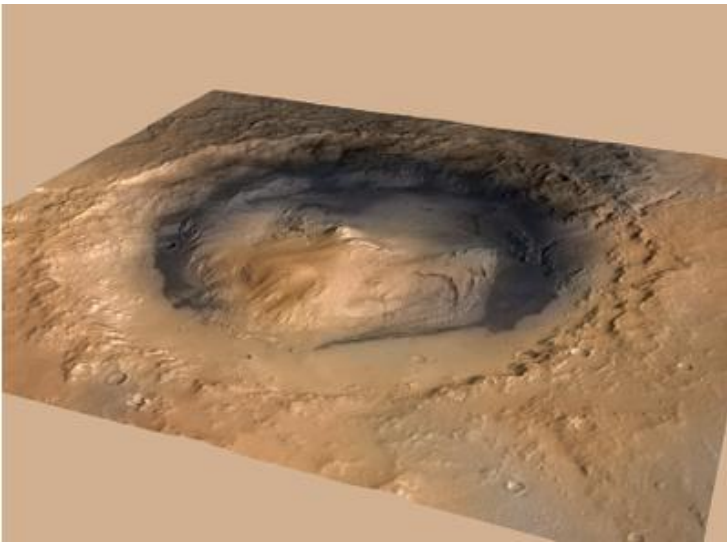


Page 3: "Analysis of the first drilled sample, from a rock target called 'John Klein,' provided the evidence of conditions favorable for life in Mars' early history: geological and mineralogical evidence for sustained liquid water, other key elemental ingredients for life, a chemical energy source, and water not too acidic or too salty."

15. Why was Gale Crater chosen over dozens of other sites, and how did the payload directly address the habitability question?

Detailed Answer:

The bottom image on page 1 shows Gale Crater's immense scale (Connecticut and Rhode Island combined). Over 100 scientists evaluated more than 30 candidate sites through multiple workshops. Gale won because Mount Sharp's layered stratigraphy records Mars' environmental transition across billions of years—essentially a geological history book. The crater floor promised clay minerals from wet periods while upper Mount Sharp layers would show the transition to the current arid state. The payload directly matched this scientific opportunity: Sample Analysis at Mars could read isotopic signatures of atmospheric loss, CheMin could identify hydrated minerals from different climate epochs, and the radiation detector could assess whether habitable subsurface environments might exist today. The site selection and instrument suite formed an integrated strategy to answer "when did Mars lose habitability, and why?"

Direct evidence from PDF:

Page 1-2: "Selection of Gale Crater followed consideration of more than 30 Martian locations by more than 100 scientists participating in a series of open workshops. The selection process benefited from examining candidate sites with NASA's Mars Reconnaissance Orbiter..."

Page 2: "In the low layers of Mount Sharp during the rover's extended mission, researchers anticipate finding further evidence about habitable past

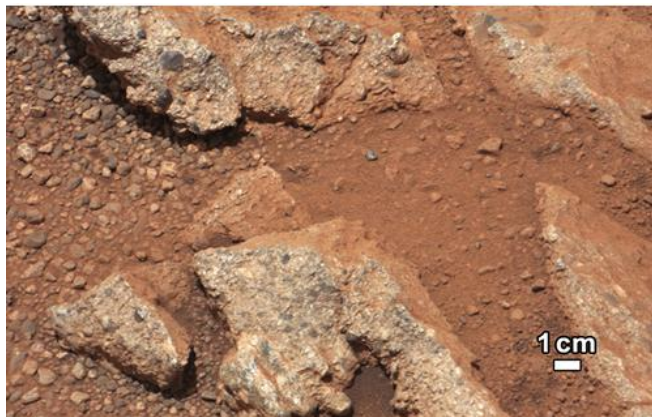
environments and about the evolution of the Martian environment from a wetter past to a drier present."

16. Explain the counterintuitive mechanism by which Mars lost most of its atmosphere from the top rather than the surface.**Detailed Answer:**

The Link outcrop image (page 2, top left) shows water-deposited rocks, evidence that Mars once had a thick atmosphere supporting liquid surface water. However, atmospheric analysis revealed Mars lost its atmosphere primarily through upper-atmospheric escape rather than surface chemical reactions (like CO₂ reacting with rocks). This occurs because Mars lacks a strong magnetic field: solar wind particles directly strike the upper atmosphere, ionizing molecules and accelerating them to escape velocity. This "top-down" loss mechanism is counterintuitive because it means surface rocks preserve evidence of ancient water while the protective atmosphere that enabled that water was being stripped away from above. The process explains why Mars transitioned from wet to dry: not

because water chemically locked into minerals, but because atmospheric pressure dropped below the triple point where liquid water can exist.

Direct evidence from PDF:



Page 2: "Analysis of Mars' atmospheric composition early in the mission provided evidence that the planet has lost much of its original atmosphere by a process favoring loss from the top of the atmosphere rather than interaction with the surface."

17. How does this single satellite image distinguish between active fire, burned area, and smoke plume?

Detailed Answer:

The Wallow Fire satellite image (page 5, bottom) uses multi-spectral analysis to differentiate fire components. VIIRS detects active fire through thermal infrared channels that sense extreme heat signatures invisible to the human eye. The charred land appears in distinct colors (shown as red outlined area in the image) because burned vegetation reflects light differently than healthy vegetation—specifically, it has altered reflectance in the near-infrared and visible red bands. Smoke plumes are identified through aerosol detection channels that measure how particles scatter light. This tri-part discrimination is critical for disaster response: active fire location guides firefighting resources, burned area extent informs damage assessment, and smoke plume tracking supports air quality warnings and aviation safety.

Direct evidence from PDF:



Image: Page 5, bottom - Wallow Fire satellite view showing charred land and smoke plumes

Page 6: "VIIRS data will also be used to identify immediate disaster effects and recovery processes from such events as fires, volcanoes, floods, hurricanes and blizzards."

18. What hemispheric asymmetry does this image reveal about climate change impacts, and why was extending the MODIS record with VIIRS scientifically essential?

Detailed Answer:

The vegetation productivity map (page 4, center) shows increased greening in the Northern Hemisphere contrasted with decreased productivity in the Southern Hemisphere—a striking asymmetry driven by different climate change mechanisms. The Northern Hemisphere trend reflects a lengthened growing season as warming pushes spring earlier and fall later at high latitudes. Meanwhile, the Southern Hemisphere shows widespread productivity decline from intensified drought over the past decade. Extending the MODIS record with VIIRS is scientifically essential because these are decadal-scale trends—they only become statistically significant and distinguishable from natural variability through continuous multi-decade observation. Breaking the data record would make it impossible to determine whether observed changes represent temporary fluctuations or permanent climate shifts.

Direct evidence from PDF:

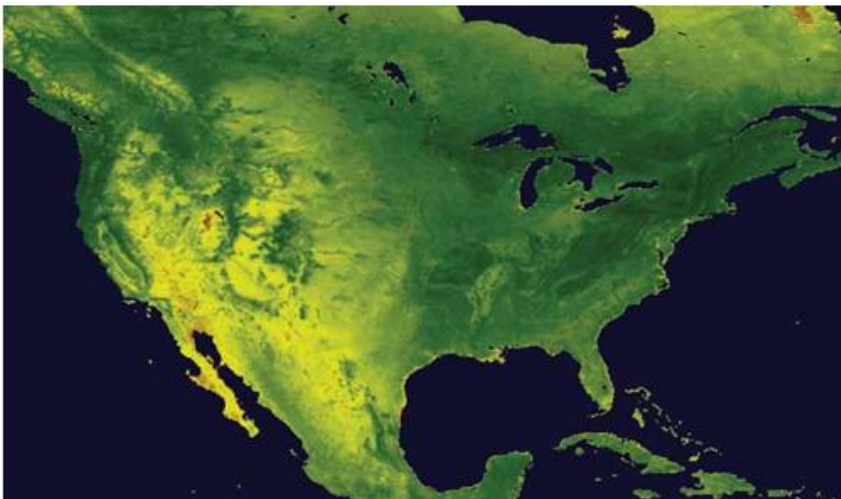


Image: Page 4, center - Vegetation across the United States using Normalized Difference Vegetation Index (NDVI)

Page 4: "VIIRS will ensure continued monitoring and analysis of recently observed decadal-scale trends of increased vegetation productivity in the northern hemisphere when compared to the southern hemisphere."

19. Why is measuring both reflected sunlight and emitted heat essential for understanding climate feedbacks, and how do clouds complicate this measurement?

Detailed Answer:

CERES measures both incoming solar radiation reflected back to space and outgoing thermal (heat) radiation emitted by Earth—together these constitute the planet's energy budget. Measuring both is essential because climate feedbacks can push these in opposite directions: for example, melting Arctic ice reduces reflected sunlight (warming feedback) while simultaneously allowing more heat emission from exposed dark ocean (cooling feedback). Clouds dramatically complicate this measurement because their effects vary by type: high, thin cirrus clouds trap outgoing heat (warming effect) while reflecting relatively little sunlight, whereas low, thick water clouds reflect substantial sunlight (cooling effect) while blocking less outgoing heat. This dual nature makes cloud

feedback one of the largest uncertainties in climate models—requiring the precise, long-term measurements CERES provides.

Direct evidence from PDF:



Image: Page 7, bottom-left - CERES instrument photo

Page 7-8: "CERES focuses exclusively on the balance of sunlight and heat in the Earth system and its change over time."

Page 3: "Some clouds reflect sunlight, keeping solar radiation from heating the Earth, yet others acts like a blanket, trapping heat near the Earth's surface."

20. Why does the ability to detect moonlight-illuminated clouds at night represent a major advance over previous imagers?

Detailed Answer:

The day/night band capability represents a major advance because it provides, for the first time, calibrated nighttime visible imagery—previous operational satellites could only image clouds at night using thermal infrared, which cannot distinguish cloud type, thickness, or texture the way visible light can. Moonlight-illuminated cloud detection enables forecasters to track fog (a major aviation and highway hazard), monitor the development of nocturnal storms, and observe low clouds that are nearly invisible in infrared but clearly visible in reflected moonlight. The calibration is crucial: unlike uncalibrated sensors, VIIRS can quantitatively measure how much light clouds reflect even under varying lunar illumination, making nighttime observations scientifically comparable to daytime data rather than just qualitative imagery.

Direct evidence from PDF:



Image: Page 6, bottom - Gulf of Mexico oil spill showing detection capabilities

Page 6: "VIIRS also has a day/night band to detect low levels of visible/near-infrared radiance at night from sources on or near the Earth's surface, such as low clouds, fog and snow cover illuminated by moonlight. This marks a significant advance over current imagers ability to aid recovery efforts."

21. Why was NPP designed as a 'bridge' mission, and what specific data continuity risks did it mitigate?

Detailed Answer:

NPP was designed as a bridge between NASA's aging Earth Observing System (EOS) satellites (Terra launched 1999, Aqua 2002, Aura 2004) and NOAA's next-generation Joint Polar Satellite System (JPSS). The specific continuity risks were severe: critical climate data records spanning decades were at risk of termination as EOS satellites aged past their design life, yet JPSS satellites weren't ready. NPP carried next-generation instruments (VIIRS, CrIS, ATMS) into orbit to: (1) test new sensor designs before committing them to operational JPSS satellites, (2) maintain unbroken data records essential for climate trend detection (gaps would make it impossible to distinguish trends from instrument calibration differences), and (3) provide real-world validation that these advanced instruments could meet both weather forecasting and climate research requirements.

Direct evidence from PDF:



Image: Page 1, center - NPP satellite illustration showing polar orbit path
Page 2: "NPP monitors the health of Earth from space—providing continuity to decades-long records and setting the stage for future Earth science missions."

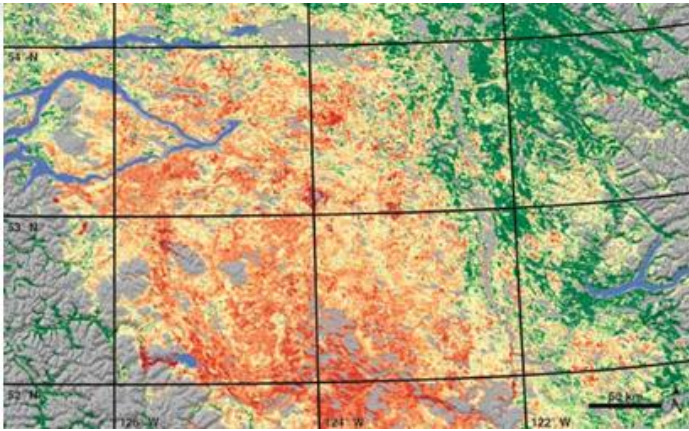
22. How does a temperature increase of just a few degrees trigger cascading ecological effects visible from space?

Detailed Answer:

The pine beetle damage image (page 4, bottom-right) illustrates how small temperature increases trigger cascading ecological collapse. Historically, severe winter cold killed pine beetle larvae, limiting infestations. A warming of just 2-3°C allows larvae to overwinter successfully, causing exponential population growth. Milder winters also stress trees through altered snowpack and soil moisture, weakening their ability to produce defensive resin. The result: beetles overwhelm tree defenses across millions of acres. This cascade becomes visible from space through dramatic vegetation index changes—MODIS detects the shift from healthy green forest (high near-infrared reflectance) to dead red-brown forest (low near-infrared reflectance). The spatial extent—entire mountain ranges turning red—demonstrates how seemingly modest temperature changes can fundamentally reorganize ecosystems at continental scales.

Direct evidence from PDF:

Image: Page 4, bottom-right - Pine beetle damage map showing red (severe damage) and green (no damage) areas in British Columbia



Page 4, caption: "Milder winter temperatures and drier summers are among the factors in increasing rates of pine beetle infestation in North America."

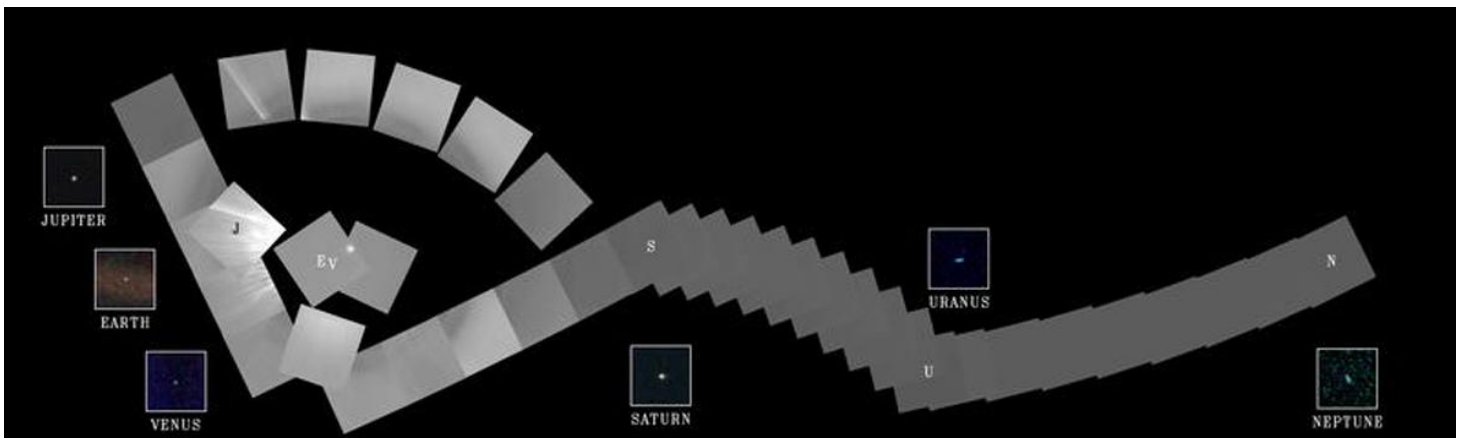
Page 3: "A change of a degree or two may seem small, but incremental increases in the overall temperature of the Earth's surface can have profound effects."

23. What fundamentally different philosophical statements do these two images from opposite directions make about humanity's place in the universe?

Detailed Answer:

The Voyager Family Portrait (page 4 of Voyager document, bottom image) and NPP's Earth-observing mission represent opposing but complementary philosophical perspectives. Voyager's image, taken from 6 billion kilometers away showing Earth as a "pale blue dot," makes a statement of cosmic humility—it reveals our planet as an infinitesimal speck in an incomprehensibly vast universe, emphasizing our smallness and fragility. This outward gaze says: "Look how insignificant we are in the cosmos." In contrast, NPP's continuous Earth observation from 824 km altitude (page 4 of NPP document, orbital diagram) makes a statement of planetary responsibility—it says our home world, though small in cosmic terms, contains everything we know and love, and deserves meticulous monitoring and care. The inward gaze says: "Look how precious and complex our home is." Together, they form a complete worldview: we are cosmically small yet locally responsible, simultaneously humbled by the universe's scale and empowered to understand and protect our world.

Direct evidence from PDFs:



Voyager page 4, bottom: Family portrait showing Earth as tiny pale blue dot among planets



NPP page 4, center: NPP satellite illustration showing polar orbit observing Earth

Voyager page 4: "In a mosaic of 60 images, it was able to capture a 'family portrait' of six of the solar system's planets, including a pale blue dot called Earth."

NPP page 2: "NPP monitors the health of Earth from space—providing continuity to decades-long records and setting the stage for future Earth science missions."

24. Why did proving past habitability on Mars require physical sampling while inferring possible present habitability on Europa relied primarily on remote visual evidence?

Detailed Answer:

The epistemological requirements differed fundamentally due to what each mission sought to prove. Curiosity's habitability claim (Curiosity page 2, John Klein drill site, bottom-right image) required demonstrating five specific chemical and geological conditions: sustained liquid water, key elemental ingredients (C, H, N, O, P, S), chemical energy sources, appropriate pH, and preservation potential. This demanded physical sampling—no remote instrument could determine clay mineral composition, measure organic chemistry, or establish that water was "not too acidic or too salty" from orbit. In contrast, Europa's habitability inference (Voyager page 3, left montage showing Jupiter and moons) relied on a single observation: a geologically young, cracked ice surface suggesting a subsurface liquid ocean. This visual evidence alone was sufficient to propose habitability potential because liquid water is life's primary requirement. Voyager couldn't drill Europa, but the visual pattern of surface fractures and few impact craters provided compelling evidence that liquid water exists beneath—making it a candidate for life without requiring in-situ chemistry.

Direct evidence from PDFs:



Curiosity page 2, bottom-right: Drill holes in John Klein rock showing powdered sample collection



Voyager page 3, left: Montage of Jupiter and its four largest moons including Europa

Curiosity page 2: "Analysis of the first drilled sample, from a rock target called 'John Klein,' provided the evidence of conditions favorable for life in Mars' early history: geological and mineralogical evidence for sustained liquid water, other key elemental ingredients for life, a chemical energy source, and water not too acidic or too salty."

Voyager page 2: "The moon Europa has a geologically young surface made of ice possibly floating on an ocean of liquid water."

25. What do these three artifacts reveal about different philosophies of designing for extremely long-duration exploration?

Detailed Answer:

The Golden Record (Voyager page 5, both images), Curiosity's RTG (Curiosity page 3, text on radioisotope power), and NPP's data continuity strategy (NPP page 1, bridge mission concept) embody three distinct long-duration design philosophies. The Golden Record represents "design for immortality"—it's built to outlast the mission by millennia, preserving human culture long after Voyager ceases functioning in ~2025. Its gold-plated copper construction and analog technology (grooves, not digital bits) were chosen because they degrade slowly in the vacuum of space over tens of thousands of years. Curiosity's RTG represents "design for predictable decline"—engineers accepted that power would decrease 9% over two years (110W to 100W), but this predictable decay allowed confident mission planning decades ahead without dependency on environmental factors like Martian dust storms. NPP's philosophy is "design for continuity through replacement"—no single satellite lasts forever, so the strategy is overlapping missions (EOS → NPP → JPSS) where each new satellite tests next-generation instruments while maintaining unbroken data records. Three philosophies: eternal preservation, graceful degradation, and planned succession.

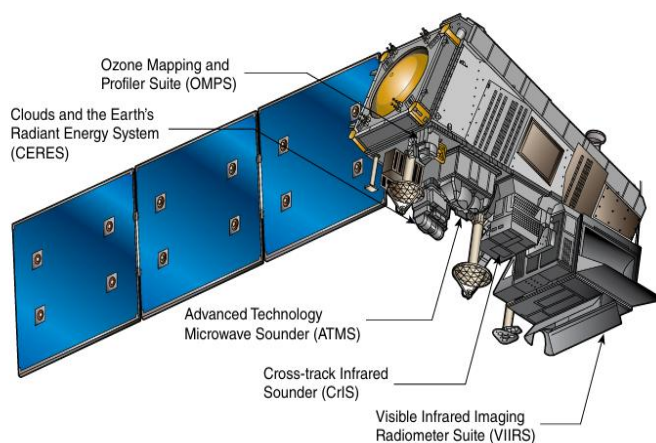
Direct evidence from PDFs:

Voyager page 5, bottom: Both sides of Golden Record showing encoded information





Curiosity page 3, center: Self-portrait showing
RTG-powered rover



NPP page 2, center: Satellite showing instrument suite
designed for continuity

Voyager page 5: "In their current paths, neither Voyager will come within less than a light-year from any star in the next tens of thousands of years, long after they will have ceased to communicate with Earth. But, just in case any civilization is able to capture them to determine their origin, each Voyager is equipped with a golden record."

Curiosity page 3: "At launch, the generator provided about 110 watts of electrical power... Although the total power from the generator will decline over the course of the mission, it was still providing more than 100 watts two years after landing."

NPP page 2: "A critical next step in this transition is the NPOESS Preparatory Project (NPP)... NPP monitors the health of Earth from space—providing continuity to decades-long records and setting the stage for future Earth science missions."